

Python 小程序 1

以此代码演示了面向对象编程、字典列表操作、异常处理和交互式菜单系统的综合应用

```
class Student:
```

```
    """学生类，封装学生信息和成绩"""
```

```
    def __init__(self, student_id: int, name: str):
```

```
        self.student_id = student_id
```

```
        self.name = name
```

```
        self.scores = {}
```

```
    def add_score(self, subject: str, score: int) -> bool:
```

```
        """添加成绩"""
```

```
        if 0 <= score <= 100:
```

```
            self.scores[subject] = score
```

```
            return True
```

```
        return False
```

```
    def get_average_score(self) -> float:
```

```
        """计算平均分"""
```

```
        if not self.scores:
```

```
            return 0.0
```

```
        return sum(self.scores.values()) / len(self.scores)
```

```
    def __str__(self) -> str:
```

```
        """学生信息字符串表示"""
```

```
        avg = self.get_average_score()
```

```
        return f"学号: {self.student_id}, 姓名: {self.name}, 平均分: {avg:.2f}, 各科成绩: {self.scores}"
```

```
class ScoreManager:
```

```
    """成绩管理系统类"""
```

```
    def __init__(self):
```

```
        self.students = {}
```

```
    def add_student(self) -> None:
```

```
        """添加学生"""
```

```
        try:
```

```
            student_id = int(input("请输入学生学号: "))
```

```
            if student_id in self.students:
```

```
                print("该学号已存在!")
```

```
                return
```

```
            name = input("请输入学生姓名: ")
```

```
            self.students[student_id] = Student(student_id, name)
```

```
            print(f"学生 {name} 添加成功!")
```

```
        except ValueError:
```

```

        print("学号必须是数字！")
def add_score_to_student(self) -> None:
    """给学生添加成绩"""
    try:
        student_id = int(input("请输入学生学号："))
        if student_id not in self.students:
            print("该学生不存在！")
            return
        subject = input("请输入科目名称：")
        score = int(input("请输入成绩："))
        if self.students[student_id].add_score(subject, score):
            print("成绩添加成功！")
        else:
            print("成绩必须在 0-100 之间！")
    except ValueError:
        print("输入格式错误！")
def show_all_students(self) -> None:
    """显示所有学生信息"""
    if not self.students:
        print("暂无学生信息！")
        return
    print("\n" + "="*50)
    print("所有学生信息:")
    for student in self.students.values():
        print(student)
    print("="*50)
def search_student(self) -> None:
    """查询学生信息"""
    keyword = input("请输入学生学号或姓名：")
    found = False
    print("\n 查询结果:")
    for student in self.students.values():
        if keyword in str(student.student_id) or keyword in student.name:
            print(student)
            found = True
    if not found:
        print("未找到匹配的学生！")
def run(self) -> None:
    """运行成绩管理系统"""

```

```
while True:
    print("\n===== 学生成绩管理系统 =====")
    print("1. 添加学生")
    print("2. 添加成绩")
    print("3. 显示所有学生")
    print("4. 查询学生")
    print("0. 退出系统")
    try:
        choice = int(input("请输入您的选择: "))
        if choice == 0:
            print("感谢使用, 再见! ")
            break
        elif choice == 1:
            self.add_student()
        elif choice == 2:
            self.add_score_to_student()
        elif choice == 3:
            self.show_all_students()
        elif choice == 4:
            self.search_student()
        else:
            print("无效的选择, 请重新输入! ")
    except ValueError:
        print("请输入数字! ")

if __name__ == "__main__":
    manager = ScoreManager()
    manager.run()
```